Building Diagnostic Robotics

5-Year Capital Plan

Sample Business Park

Building Diagnostic Robotics

Annual localized-repair allocations and modeled section outcomes.

Building
Sample Business Park

Location
123 ABC St, Brooklyn, NY

Assessment Date
May 13th, 2026 10:00 AM

Planning Period
2026–2030

Headline Findings

$18,847	$4,000	$1,768
Program Investment	Annual Budget	Deferred Priced Scope

Roof Life Gained: **+2.1 years**

Roof-Life Value Gain: **$18,097**

1 Executive Summary

The plan evaluates Sample Business Park's 3 roof sections covering 5,483 sq ft. It identifies **$20,615** of priced, localized repair scope and schedules **$18,847** within the 5-year funding horizon. The scheduled program is modeled to increase the roof's area-weighted remaining life from **3.4** to **5.5** years remaining.

If no action is taken, the roof's current condition is **Fair** and has an estimated **3 yr 4 mo** of remaining service life.

The plan prioritizes modeled section urgency and high-severity repairs within annual funding; it does not authorize construction or replace contractor validation.

2 5-Year Plan

2.1 Plan at a Glance

Year	Program Focus	Budget	Scheduled	Carry-forward
2026	High-severity stabilization: flashing and membrane repair items	$4,000	$3,985	$15
2027	Membrane and flashing repair items	$4,000	$3,946	$70
2028	Membrane and flashing repair items	$4,000	$4,013	$57
2029	Membrane and flashing repair items	$4,000	$2,972	$1,084
2030	Membrane and flashing repair items	$4,000	$3,932	$1,153
Program totals		**$20,000**	**$18,847**	**$1,153**

Table 2.1. 5-year program summary

2.2 Year 1 (2026)

In 2026, the program focuses on high-severity stabilization: flashing and membrane repair items across Section 1, Section 2, and Section 3. It funds 8 planned repair items for $3,985 of $4,000 available funding and carries $15 forward.

Repair Item	Section	Severity	Quantity	Cost
Base Flashing Membrane Material	1	Medium	5 LF	$42
Flashed Penetration	1	High	1 EA	$83
Base Flashing	3	Medium	10 LF	$85
Base Flashing	2	High	80 SF	$2,009
Metal Cap Flashing	2	High	24 SF	$603
Metal Cap Flashing	2	Medium	40 LF	$339
Patching	2	Medium	175 SF	$657
Splits	2	High	25 SF	$167
Year 1 total (available funding: $4,000)				**$3,985**

Table 2.2. Year 1 repair scope

2.3 Year 2 (2027)

In 2027, the program focuses on membrane and flashing repair items across Section 2. It funds 4 planned repair items for $3,946 of $4,015 available funding and carries $70 forward.

Repair Item	Section	Severity	Quantity	Cost
Base Flashing	2	Medium	140 LF	$1,186
Defective Seams	2	Low	10 LF	$145
Defective Seams	2	Medium	25 LF	$362
Patching	2	Low	600 SF	$2,254
Year 2 total (available funding: $4,015)				**$3,946**

Table 2.3. Year 2 repair scope

2.4 Year 3 (2028)

In 2028, the program focuses on membrane and flashing repair items across Section 1. It funds 4 planned repair items for $4,013 of $4,070 available funding and carries $57 forward.

Repair Item	Section	Severity	Quantity	Cost
Flashed Penetration	1	Medium	3 EA	$249
Improper Equipment Supports	1	Low	27 EA	$3,240
Patching	1	Low	100 SF	$277
Ridges	1	Low	15 SF	$247
Year 3 total (available funding: $4,070)				**$4,013**

Table 2.4. Year 3 repair scope

2.5 Year 4 (2029)

In 2029, the program focuses on membrane and flashing repair items across Section 1 and Section 3. It funds 7 planned repair items for $2,972 of $4,057 available funding and carries $1,084 forward.

Repair Item	Section	Severity	Quantity	Cost
Debris Vegetation	1	Low	4 EA	$20
Defective Seams	1	Medium	15 LF	$217
Base Flashing	3	Low	25 LF	$212
Debris Vegetation	3	Low	2 EA	$10
Defective Seams	3	Low	5 LF	$72
Patching	3	Low	575 SF	$2,160
Patching	3	Medium	75 SF	$282
Year 4 total (available funding: $4,057)				**$2,972**

Table 2.5. Year 4 repair scope

2.6 Year 5 (2030)

In 2030, the program focuses on membrane and flashing repair items across Section 2. It funds 3 planned repair items for $3,932 of $5,084 available funding and carries $1,153 forward.

Repair Item	Section	Severity	Quantity	Cost
Base Flashing	2	Low	110 LF	$932
Metal Cap Flashing	2	Low	45 LF	$381
Ridges	2	Low	150 SF	$2,619
Year 5 total (available funding: $5,084)				**$3,932**

Table 2.6. Year 5 repair scope

3 Deferred Scope and Funding Review

Deferred items were not funded within the configured planning horizon or require pricing before they can be evaluated for funding. They remain visible for future funding and contractor-review decisions.

Repair Item	Section	Severity	Amount
Moisture Damage	1	Low	Not priced
Improper Equipment Supports	3	Low	$600
Moisture Damage	3	Low	Not priced
Debris Vegetation	2	Low	$155
Moisture Damage	2	Low	Not priced
Ponding	2	Low	$684
Surface Deterioration	2	Low	$329
Deferred priced scope			**$1,768**

Table 3.1. Deferred scope and funding review

4 Modeled Section Outcomes

Modeled outcomes reflect funded repair items with reliable quantity mappings. They are planning estimates, not guarantees of service life, roof performance, or contractor results.

Section	Initial Condition	Final Condition	Roof Life Added	Funded Work
1	Good (RCI 57.1)	Good (RCI 61.5)	+0.8 years	$4,375
3	Good (RCI 61.9)	Good (RCI 66.3)	+0.9 years	$2,820
2	Poor (RCI 35.9)	Good (RCI 67.6)	+5.6 years	$11,652

Table 4.1. Modeled section outcomes